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JEMS 8 (2019), 1: 117–139

Common Conceptions and the Metaphysics of Material Substance:Domingo de Soto, Kenelm Digby and Johannes de Raey

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Abstract: This paper explores how, according to three early modern philosophers, philosophical theory should relate to our pre-theoretical picture of reality. Though coming from very different backgrounds, the Spanish scholastic, Domingo de Soto, and the English natural philosopher, Kenelm Digby, agreed that an ability to accommodate our pre-theoretical picture of the world and our ordinary way of speaking about reality is a virtue for a philosophical theory. Yet at the same time, they disagreed on what kind of ontology of the material world is implied by these. The Dutch Cartesian, Johannes de Raey, took a very different approach, and argued that the picture of reality we naturally develop from our early days onwards and the language associated with it have their use in domains such as law and medical practice, but are a poor guide to the ontology of the material world. Thus, if we are to arrive at a proper understanding of the nature of matter, we need to move beyond the picture of reality we naturally develop from our early days onwards in order to come to see that the nature of matter consists in bare extension.

Keywords: Domingo de Soto, Kenelm Digby, Johannes de Raey, common notions, real accidents, prime matter

How should philosophy relate to our pre-theoretical picture of reality? On one view, philosophy takes its cues from this picture, and tries to spell out in detail what it has to say about good and evil, knowledge, and the nature of the ordinary objects we see around us. On a competing view, the very point of philosophy is to move beyond our pre-theoretical picture of reality, and to point out where it stands in need of correction. By looking at three case studies, this paper will illustrate how both of these views can be found in early

modern discussions of the ontology of material reality. These case studies will not only deepen our knowledge of the role played by appeals to what was often called our common or natural picture of reality, but will also remind us of a philosophical problem that such appeals face.

The first author we will look at is the Spanish Dominican, Domingo de Soto.[1] Though little known today, Soto was the author of an influential commentary on the *Sentences* as well as an important treatise on the Aristotelian categories. One of the issues he addressed in these works was the ontological status of the accidents of material substances. Are these accidents real beings over and above the substances whose accidents they are, as realists claim? Or do accidents somehow reduce to substances, as nominalists submit? As we will see, Soto argued that at least some accidents are real and have a being of their own over and above that of their bearers. This kind of realism was of course not new with Soto. But what makes his defence of realism of interest for present purposes, is the way in which it appeals to what he called our common way of conceiving of things. According to Soto indeed, our common way of conceiving of reality suggests a realist view of at least some accidents.

The second author we will look at is the English courtier and philosopher, Sir Kenelm Digby.[2] An early member of the Royal Society, Digby has been described as one of the most influential philosophers of his time[3] and as the first to offer a full statement of an atomist system of natural philosophy in the English language.[4] But even if he was in many ways at the forefront of the new philosophy of the century, Digby at the same time remained firmly attached to the legacy of Aristotle, which he believed could well go hand in hand with the innovations that were taking place in the science and philosophy of his time.

One point where Digby took himself to agree with Aristotle, was that the bodies we see around us are best described as substances endowed with accidents of nine sorts. This view naturally led him to consider the ontological status of the accidental categories. As we will see, Digby was critical of category realism here, and argued that it was in part the product of an artificial way of describing material reality. The natural picture of reality common to all men,

[1] On the career and work of Soto (1494–1560), see David Jiménez Castaño, "Domingo de Soto. Una breve bibliografía para iniciarse en el estudio de su obra y de su pensamiento," *Azafea. Revista de filosofia* 18 (2016), pp. 113–128.

[2] On the life and work of Digby (1603–1655), see Vittorio Gabrieli, *Sir Kenelm Digby. Un inglese italianato nell'età della controriforma*, Rome: Edizioni di Storia e Letteratura, 1957.

[3] Antonio Clericuzio, *Elements, Principles and Corpuscles*: *A Study of Atomism and Chemistry in the Seventeenth Century*, Dordrecht: Kluwer, 2000, p. 81.

[4] On Digby's contribution to atomism in England see Robert Kargon, *Atomism in England from Harriot to Newton*, Oxford: Oxford University Press, 1966, pp. 66–67.

he believed, suggested an ontology in which accidents are not beings in their own right over and above the substances whose accidents they are.

The cases of Soto and Digby are telling in at least two ways. First, the case of Soto reminds us that, for all the criticism they would later receive, doctrines such as category realism were at least in part the result of an attempt to spell out in detail the ontological commitments of what was seen to be a natural and common way of conceiving of reality. Second, the cases of Soto and Digby raise a philosophical question about the weight appeals to this way of conceiving can carry. For the moment we find one author appealing to the common way of conceiving of things to defend a form of category realism and another to deny it, the notion of a natural or common way of conceiving of the material world begins to look rather flexible indeed. In fact, it begins to look doubtful whether there are any truly natural and common views of the ontological status of accidents at all, over and above the views common to those in the nominalist camp and those in the realist camp already. If the answer to this question is no, the philosophical value of appeals to a common or natural way of conceiving of the world seems to be limited at best.

This at any rate appears to have been the view of one of the most eminent Dutch Cartesians of the seventeenth century, Johannes de Raey.[5] As we will see, de Raey believed that the task of philosophy is not so much to build on, as to replace our pre-theoretical picture of reality. Our pre-theoretical ideas about the way the world is and the language in which it is typically couched, de Raey maintained, are useful tools to navigate the demands of everyday life. But they are a burden to progress in the theoretical disciplines of science and philosophy.

1. *Soto on Accident Realism and the Common Way of Conceiving of Material Reality*

According to Soto, we describe the material world in terms of the Aristotelian ten categories. Material reality is populated by substances endowed with accidents falling under nine broad categories, that is. But what is the

[5] On de Raey (1622–1702) in the context of Dutch Cartesianism, see Tad Schmaltz, *Early Modern Cartesianisms: Dutch and French Constructions*, Oxford: Oxford University Press, 2017, pp. 77–83. According to de Raey, to follow Descartes need not be to turn away from Aristotle. As he argued in several of his works, the philosophy of Aristotle had been perverted by his scholastic readers, and the difference between Aristotle and Descartes was often smaller than that between Aristotle and the scholastics. See Christia Mercer, *Leibniz's Metaphysics: Its Origins and Developments*, Cambridge: Cambridge University Press, 2001, pp. 107–109 for discussion. Digby and de Raey would likely have agreed on this point. Yet as we will see towards the end of this paper, they at the same time disagreed on fundamental points of methodology as well as doctrine.

ontological status of these accidents? Are they beings in their own right that are somehow added to substances? Or do they somehow reduce to substances and their material parts? This traditional question about the ontological foundation of the Aristotelian categorization of reality was taken up in detail by the Dominican philosopher and theologian, Domingo de Soto, in his *Sentences* commentary.

There, Soto argues that the case for accidents in the categories of Quantity and Quality can be settled by considering the miracle of the Eucharist.[6] In the Eucharist, the host undergoes a transformation as it ceases to be a piece of bread and becomes the body of Christ. But even though the piece of bread thus ceases to exist, its bready accidents do not. We still perceive the colour and taste of bread, for instance. And this raises the question of what these accidents inhere in. After all, the bread in which they used to inhere has ceased to exist, and has become the body of Christ. But being a human body, that is not a suitable bearer for the colour and taste of bread. Thus after the consecration of the host, its bready accidents seem to somehow exist on their own, without a suitable substance to bear them.

Soto accepted this outcome, arguing that material substances such as a piece of bread have a layered ontological structure. Thus the quantity of the bread inheres directly in the material substance, causing its part to be spread out in space. Other accidents such as its colour and taste in their turn inhere in the quantity of the bread. This analysis has two advantages, Soto believed. First of all, it provides a direct explanation of why accidents such as colour and taste are spread over the whole of the bread: they are so spread out, because they inhere in the bread via its quantity or extension. Second, it provides a way of accounting for the sacrament of the Eucharist. Indeed, Soto held that, after the consecration of the host, the quantity of the bread comes to serve as a kind of substance to bear the other accidents, which continue to inhere in quantity as before.

Thus with quantity, we have at least one real accident: an accident that has a being in its own right over and above the substance whose accident it is. But Soto takes his commitment to real accidents a step further and argues that, if God can sustain quantity apart from its substance, there is no reason why he could not perform the same kind of miracle for accidents in the category of quality. Thus Soto arrives at the conclusion that, if God so wishes, even accidents in the category of quality can "truly exist on their own," and so "acquire the mode of being of substances."[7] Just as accidents in the category of quantity, then, accidents in the category of quality are real in the sense that they

[6] See Domingo de Soto, *Commentarii in Quartum Sententiarum*, Salamanca: Apud Ioannem Mariam a Terranova, 1569, d. 11, q. 1, aa. 1–2 (esp. pp. 434–436).

[7] "*Vero per se substunt virtute divina, sic ut per consecrationem acquirant quodammodo illud esse, ut ad modum substantiae sint.*" *In Quartum Sententiarum*, d. 11, q. 1, a. 1 (p. 434).

resist reduction to, and do not depend for their existence on, the substances in which they naturally inhere.

Soto addresses the ontological status of accidents in his commentary on the *Categories* as well. There, he briefly summarizes the Eucharistic argument for the reality of at least certain accidents, but also introduces a new line of argument for quantity realism. According to Soto indeed, quantity nominalism sits uneasily, not only with Aristotelian doctrine, but also with our common way of conceiving of things:

> I cannot approve of the way of speaking of some Nominalists on this score; namely those who concede propositions like the following: a substance is a quantity . . . Not that I would go so far as to say they must be rejected altogether, but they certainly seem to fly in the face, not only of Aristotelian doctrine, but also of the common way of conceiving of things.[8]

What does Soto mean by the common way of conceiving of things here? A first option could be to interpret this as a reference to what many scholastics, following Boethius, called common conceptions of the mind.[9] These *communes animi conceptiones* were taken to be principles whose truth is immediately evident to those who understand the terms of which they are composed, such as the principle that the whole is greater than its part, or the principle that when equal amounts are taken away from equal amounts, equal amounts remain.[10]

Yet it does not seem likely that, in speaking of the common way of conceiving of things, Soto was referring to principles of this kind. For in his discussion of quantity, Soto never refers to the kind of principles identified by Boethius, and when he claims that quantity nominalism sits uneasily with the common way of conceiving of things, there is no evidence that he intends this as a claim that quantity nominalism somehow contradicts a self-evident principle.

Moreover, by the time Soto was writing, *communes animi conceptiones* had become the standard term to denote principles whose truth is evident from the terms that compose them. Soto does not use this term. The term he does

[8] "*Modum loquendi aliquorum Nominalium satis in hac parte probare non possum: sane qui huiusmodi propositiones concedunt, substantia est quantitas . . . Non quod ego sim tanti, ut contendam eas omnino repudiare: sed certe quod videntur pugnare, non solum cum doctrina Aristotelis, sed etiam cum communi modo concipiendi.*" Domingo de Soto, *In Porphyrii Isagogen, Aristotelis Categorias, librosque de Demonstratione*, Venice: ex Officina Dominici Guerraei, 1587, *In Porphyrii*, De Quantitate, q. 2 (p. 181).

[9] For some discussion, see Edward Grant, *God and Reason in the Middle Ages*, Cambridge: Cambridge University Press, 2001, pp. 43–44.

[10] Aquinas offers a detailed discussion of Boethius on common conceptions of the mind in the first chapter of his *Expositio libri De ebdomadibus*. See also Thomas Aquinas, *Summa theologiae*, prima pars, q. 2, a. 1.

use, *communis modus concipiendi*, does not appear to have been a technical term, and was regularly used by late scholastic authors in a broad sense to denote commonly held beliefs, or common ways of understanding a certain word or concept.[11] Thus when Soto claims that quantity nominalism sits uncomfortably with the common way of conceiving of things, it seems plausible to read him as claiming that quantity nominalism sits uncomfortably with commonly shared beliefs about the world and the way in which the words we use to describe it are commonly understood.

To see why Soto would make this claim, consider a man and his quantity. Being a substance, the man is without doubt a real thing[12] in the world. So we have:

1. The man is a *res*.

But what about his quantity? According to the quantity nominalist, the quantity is no *res* over and above the substance. So we have:

2. His quantity is the same *res*.

But from these it follows that:

3. The man is his quantity,

or indeed that there is a substance that is a quantity. Now according to Soto, some nominalists had been happy to accept this kind of conclusion. Thus William of Ockham had submitted that the following argument must be accepted:

> Whatever is a corporeal substance, is a quantity. But the body of Christ is a corporeal substance. Therefore the body of Christ is a quantity.[13]

[11] Thus Francisco de Toledo tells us that, according to the common way of conceiving of faith, faith comes from hearing. Only on a special understanding of the word is faith the result of revelation. See his *Commentarii et annotationes in epistolam beati Pauli apostoli ad Romanos*, Rome: Sumptibus Paulini Arnolphini Lucensis, 1602, p. 524. For a similar use of the term, see Antonio Rubio, *Commentarii in universam Aristotelis dialecticam*, Cologne: Crithius, 1609, p. 140.

[12] The Latin here has *res*, which was a term often used to refer to substances or real qualities. (See also Robert Pasnau, *Metaphysical Themes. 1274–1671*, Oxford: Oxford University Press, 2011, p. 136.) In what follows, I will often leave *res* untranslated.

[13] "*Et ideo talis modus arguendi valet 'quidquid emis, comedis; carnes crudas emis; igitur carnes crudas comedis,' 'quidquid est substantia corporea, est quantitas; corpus Christi est substantia corporea; igitur corpus Christi est quantitas,' et sic de aliis.*" William of Ockham, *Summa logicae*, ed. Philotheus Boehner, New York: The Franciscan Institute, 1974, pars 3.4, cap. 10.

Ockham accepts the first premise, because he believes that all corporeal substances have a quantity and that the quantity of a material substance is not some *res* over and above it. But if every material substance has a quantity that is the same *res* as it, every material substance is a quantity. Since the second premise is obviously true, it must be granted that the body of Christ is a quantity.

But according to Soto, this way of speaking flies in the face of what we commonly believe men to be and the way in which we normally respond when asked what a man is:

> To concede that a substance is a quantity . . . goes against the common way of conceiving of things. For to someone who asks what a man is, it is answered that a man is a substance, and we do not say that he is a quantity.[14]

I take it that the "we" in the last sentence here is an inclusive one, and read Soto as claiming that no one would say that a man is a quantity, and that, when asked what a man is, all men, no matter how learned or unlearned they may be, would agree that a man is substance of some kind.

The ontology that is suggested by this answer, Soto believes, is a realist one. We do not say that a man is a quantity, because "a man and a quantity are two, not only according to the connotations of the words, but in reality as well."[15] And a similar case can be made for accidents in other categories. Take for instance accidents in the category of posture. According to a nominalist such as Ockham, accidents in this category reduce to their substances, in the sense that the sitting or standing posture of a man reduces to a certain spatial arrangement of his material parts. But if a man is a *res* and his posture is the same *res*, it will follow that a man is his sitting posture.[16] This again flies in the face of our common way of speaking and conceiving of material substances. Yet it is not easy to see, Soto believes, "how statements of this kind can be avoided by someone who does not allow that between a sitting person and his posture, there is the same distinction that obtains between two *res*."[17] Again, that is, ordinary parlance and the picture of reality it conveys seem to speak

[14] "*Concedere substantiam esse quantitatem, aut aliud caeterorum praedicamentorum sit contra commune modum concipiendi: nam quaerenti quid sit homo, respondetur, quod est substantia, et non respondemus, quod est quantitas.*" *In Porphyrii*, De Quantitate, q. 2 (p. 182).

[15] "*Non solum propter connotationes terminorum, sed re ipsa videntur esse duo, homo et quantitas.*" *In Porphyrii*, De Quantitate, q. 2 (p. 182).

[16] "*Eadem res quae est sedens, est sessio. Undo continuo sequitur, haec res est hoc (demonstrando hominem sedentem), et eadem res est sessio, ergo sedens est sessio.*" *In Porphyrii*, De Quantitate, q. 2 (p. 182).

[17] "*Dubium restat, quomodo posset huiusmodi affirmationes evitare qui non concederet inter sedentem et sessionem eam distinctionem, quae est inter duas res.*" *In Porphyrii*, De Quantitate, q. 2 (p. 182).

in favour of category realism, whereas the kind of language sanctioned by category nominalism strikes us as odd.

In all fairness, Soto does not believe this to be in any way a conclusive argument for category realism. Indeed, in his commentary on the *Categories*, he spends some time outlining a way in which realists could plausibly try to resist inferences such as (1)–(3) above,[18] and even admits to some doubt as to whether or not a body and its posture are distinct in the same way in which the size and colour of the host are distinct from it.[19] But in spite of these nuances, the overall tenor of Soto's discussion seems clear enough. Our ordinary way of conceiving things, which is reflected in the way we respond when asked what a man is, suggests a realist ontology of at least some kinds of accidents. If a nominalist wishes to resist that suggestion, he will need to do some work to accommodate ordinary parlance, and to show that he can resist having to grant that a man is his quantity, or that a sitting person is his posture.

2. *Digby on Common Notions as an Antidote against Accident Realism*

In the beginning of his *Two Treatises*, published in 1644, Digby draws a distinction between two kinds of language:

> We may observe there are two sorts of language to expresse our notions by. The one belongeth in general to all mankind, and the simplest person, that can but apprehend and speake sense, is as much iudge of it, as the greatest Doctour in the schooles expresse the thinges properly and plainely, according to the natural conceptions that all people agree in making of them. The other langue, is circled in with narrower boundes, and is understood onely by those that in a particular and expresse manner have been trained up unto it.[20]

The first language is common to all men, and expresses our pre-theoretical picture of the world. It conveys our common notions of things: the concepts

[18] In particular, he suggests that a nominalist could argue that there is an equivocation on 'is' in the premises and the conclusion of this inference, by drawing a distinction between 'is' *in sensu identico* and *in sensu formalis*. In the first sense, when we say that *x* is *y*, we say that *x* refers to something that *y* also refers to. Thus we use 'is' in this way when we say that the thing on my left is the thing on your right. But when we say *x* is *y* in the second sense, we say that *y* constitutes the essence or *ratio* of *x*. This is how we use 'is' when we say that man is a rational animal. Now according to Soto, the nominalist might argue that in (1) the man is a *res* and (2) his quantity is the same *res*, the copula is used *in sensu identico*, whereas in (3) the man is his quantity, the copula is used *in sensu formalis*. This equivocation on 'is' in the argument would make it fallacious. See *In Porphyrii*, De Quantitate, q. 2 (pp. 182–3).

[19] *In Porphyrii*, De Quantitate, q. 2 (p. 181).

[20] Kenelm Digby, *Two treatises. In the one of which, the Nature of Bodies; in the other, the Nature of Mans Soule; Is Looked into: In the Way of Discovery, of the Immortality of Reasonable Soules*, Paris: Gilles Blaizot, 1644, p. 5.

that we are naturally led to make when things first leave their impressions on our senses. The second language expresses the picture of the world that is laid down in scientific theory and philosophy. It conveys the theoretical notions used by scholars and specialists to describe the world we live in.

According to Digby, among the central concepts of the first kind of language are in fact those of the ten Aristotelian categories:

> Of the first kind, are those tenne general heads, which Aristotle calleth Praedicaments: under which he (who was the most iudicious orderer of notions, and directour of mens conceptions that ever lived) hath comprised whatsoever hath, or can have a being in nature. For when any obiect occurreth to our thoughts, we eyther consider the essentiall and fundamentall Being of it; or we referre it to some species of Quantity; or we discover some qualities in it; or we perceive that it doeth, or that it suffereth some thing; or we conceive it in some determinate place, or time, and the like. Of all which, every man living that enioyeth but the use of reason, findeth naturally within himselfe at the very first naming of them, a plaine, complete, and satisfying notion.[21]

Now when Digby says that the categorical concepts form the core of the first kind of language, part of what is saying is that it is not for the learned to determine the meaning of these concepts. On the contrary, his view is that:

> The first worke of schollers, is to learne of the people . . . what is the true meaning and signification of these primary names.[22]

Now even though ordinary people have the first word on the meaning of categorical concepts, Digby does not mean to claim that they also have the last word. There may well be more to be said about substances and their accidents than first meets the eye or than is captured in our pre-theoretical image of the world. But Digby does claim that theoretical notions of the categories may not oppose the common notions communicated by the first kind of language, which he sees as the "basis and foundation" for specialized discourse.[23]

2.1. Digby on Quality

Early on in the *Two Treatises*, Digby draws a distinction between two accounts of quality. The first is the account that "Philosophers do ordinarily

[21] *Two Treatises*, p. 5.

[22] *Two Treatises*, p. 8.

[23] *Two Treatises*, p. 8. As Andreas Blank has argued, the idea in Digby that common notions set boundaries for theoretical speculation traces back to his Parisian friend Pierre Gassendi, and ultimately to Stoic tradition. See Andreas Blank, "Composite Substance, Common Notions, and Kenelm Digby's Theory of Animal Generation," *Science in Context* 20 (2007), pp. 1–20, esp. p. 15.

make of them." The second is the concept of a quality as it is "ordinarily understood in common conversation."[24] The first notion of quality he rejects, but the second he accepts.

The philosophical notion of a quality, according to Digby, is that of a real accident: an accident that has a being of its own over and above the substance whose accident it is. Digby rejects this notion of quality, because he takes it to involve a contradiction. On the one hand, philosophers claim that qualities are real entities with a being of their own over and above that of the bodies whose qualities they are. In doing so, they imply that qualities could subsist on their own in the way of substances. But on the other, philosophers also speak of qualities as accidents, and accidents are defined as beings that cannot subsist on their own in the way of substances. So in speaking of qualities as accidents with a being of their own, they describe qualities as items that possess and lack the ability to subsist on their own at the same time:

> A reall Entity or thing, must necessarily have an *Existence* or *Being* of its owne: which they allow them. And whatsoever hath so, becometh a substance, for it subsisteth by its owne Existence . . . And then presently to say that it doth not subsist of it selfe; or that it requireth the subsistence of a substance . . . is a pure contradiction.[25]

According to Digby, the willingness of many philosophers to accept that qualities can behave like substances can be traced back, at least in part, to their disregard for such common notions as the common notion of existence:

> This ariseth from a wrong notion they make to themselves of substance, existence and subsistence: and from their not consulting sufficiently with theire owne thoughts.[26]

The diagnosis here is perhaps not as clear as one might wish, but the point seems to be as follows. Philosophy has given birth to an artificial notion of existence, according to which certain things can exist at the same time in the way of substances and in the way of accidents. This kind of existence is then ascribed to qualities, which the philosophers believe on the one hand to exist in and of themselves in the way of substances, and on the other hand to inhere in a substrate in the way of accidents. This goes against the common notion of existence, according to which an item either exists as a substance or as an accident, but not in both ways at the same time. Thus if only the philosophers had taken the time to properly consult their own natural notions, they would never have endorsed the account of quality Digby is rejecting here.

[24] *Two Treatises*, pp. 39–40.
[25] *Two Treatises*, p. 39.
[26] *Two Treatises*, p. 39.

Digby contrasts the philosophical notion of quality with the common notion of quality. According to Digby, the common notion of qualities defines them as the "particularities wherein one thing differeth from an other," or as ways of being that one substance does not share with others.[27] The best way to do justice to this, he believes, is to say that the qualities of bodies reduce to these bodies and their material constitutions. Thus "the cubicall figure of a deye, is nothing else but the very body of the deye it selfe,"[28] and the colour and smell of an apple

> are one and the same substance of the apple, that maketh various and different impressions upon me, according to the various dispositions of my severall senses.[29]

More precisely, to say that an apple is red just is to say that it has a material constitution such that it will appear as red to the human eye, to say that it is sweet just is to say that its material constitution is such that it will appear as sweet to the human sense of taste, and to say that it is cold just is to say that its material constitution is such as to make it appear cold to the human sense of touch. In brief, to ascribe certain sensory qualities to an apple is to make an ontological commitment to nothing more than a substance with a material make-up that makes it affect our senses in a certain way.[30]

2.2. Digby on Quantity

Matters are a bit more complicated for the extension of material substances, or what Digby calls their quantity.[31] In the bodies we see around us, Digby believes, there is a "reall divisibility between quantity and substance."[32] By this he does not mean that the quantity of a body could be separated from the body and continue to exist on its own. Rather, in saying that there is a real divisibility between quantity and substance, Digby means that the quantity of a body is variable.[33] And by this he means that it is possible for some given portion of matter to gain or lose quantity without gaining or losing matter.

[27] *Two Treatises*, p. 2.

[28] *Two Treatises*, p. 41.

[29] *Two Treatises*, p. 2.

[30] On Digby on bodily qualities, see also Clericuzio, *Elements, Principles and Corpuscles*, pp. 81–82.

[31] According to Digby, "Extension or divisibility is the common notion of Quantity." *Two Treatises*, p. 10.

[32] *Two Treatises*, p. 26.

[33] Digby tells us that "the one of them can be changed without the other." *Two Treatises*, p. 25.

Digby thus rejects what one scholar has recently coined the Conservation of Quantity Principle.[34] This principle says that the quantity of a portion of matter is constant, and that there can be no gain or loss of quantity without a gain or loss of matter. As Descartes, one of the most eminent proponents of this principle, put it:

> It is a complete contradiction to suppose that something should be augmented by a new quantity or new extension without new extended substance, i.e. a new body, being added to it at the same time.[35]

Digby rejects this principle for two reasons. The first is that, if the quantity of a body is to remain constant, the natural process of rarefaction becomes impossible to account for. For on the assumption that the quantity of a body remains constant, its rarefaction will have to come out as either an intrusion of alien matter between its material parts, or as a dilation or increase in number of its interparticular pores. But Digby denies that either of these accounts can be made to work, and concludes that rarefaction and condensation can only be accounted for on the assumption that the quantity of a body is variable. Rarefaction can then be described as a process in which a body inflates as it gains quantity without gaining matter, while condensation will come out as a process in which a body deflates as it loses quantity without losing matter.[36]

The second reason why Digby rejects the conservation of quantity principle, is that it flies in the face of our common notion of quantity:

> For howsoever, witty explications may seeme to evade, that the same thing is now greater, now lesser, yet it can not be avoided, but that ordinary men, who looke not into Phylosophy, do both conceive it to be so, and in theire familiar discourse expresse it so.[37]

When ordinary men see boiling water, Digby claims, they naturally come to conceive of the water as a portion of matter that can take up a larger or a smaller quantity, and of its quantity as a feature that can vary even as the matter whose feature it is remains the same. Philosophy needs to be able to

[34] Pasnau, *Metaphysical Themes*, pp. 71–76. Pasnau argues that, while the conservation of quantity principle was widespread among post-scholastic philosophers, most scholastics rejected it.

[35] René Descartes, *Œuvres de Descartes*, ed. Charles Adam and Paul Tannery, 2nd ed., 11 vols., Paris: Vrin, 1964–1974, vol. 8A, p. 44 / *The Philosophical Writings of Descartes*, ed. and trans. John Cottingham, Robert Stoothoff, Dugald Murdoch, and Anthony Kenny, 3 vols., Cambridge: Cambridge University Press, 1984–1991, vol. 1, p. 226.

[36] See *Two Treatises*, pp. 19–32. This is not the place to offer a detailed account of Digby on rarefaction and condensation. For some discussion, see Douglas M. Jesseph, *Squaring the Circle: The War between Hobbes and Wallis*, Chicago: University of Chicago Press, 1999, pp. 136–139.

37 *Two Treatises*, p. 25.

accommodate this common notion of quantity. According to Digby, the best way to do this is to say that, even though quantity never subsists apart from a body whose quantity it is, quantity remains distinct from the body whose quantity it is, in the sense that it is possible for the quantity of a body to change, even as the material out of which it is built up remains constant.

3. *Daily Practice versus Metaphysical Rigour. The Case of De Raey*

Like Digby, de Raey distinguishes between two kinds of language. The first kind of language is common to all men, the second is the language of philosophers:

> Understanding and speech are twofold. The first, it appears, is prior in nature, and is common to all men, arts and disciplines. The other is posterior by nature, and is proper to Philosophy.[38]

According to de Raey, the first kind of language communicates the pre-theoretical picture of the world that we develop on the basis of the impressions objects leave on our senses. This is a picture we begin to develop from our earliest age onwards, and this is what makes the language we use to communicate it prior in nature. The second language communicates the theoretical picture of the world we develop on the basis of our scientific and philosophical reflections, and is developed at a later point in our lives.

But where Digby believed that common language as it were set the boundaries within which specialized discourse could move, de Raey holds that the two kinds of language are best confined to their own domains:

> These two branches of understanding and speech must be kept distinct as much as possible, and as a result of this most powerful distinction, we must hold that Philosophy by its nature is distinct from any other art or discipline.[39]

In particular, the division of labour de Raey envisions is as follows. As we will see in section 3.1, the first kind of language is the language that best serves that needs of our daily lives. In the practice of everyday life, the technical nature of the second kind of language would only be a burden, de Raey

[38] "*Duplex intellectus et sermo duplex, unus ac natura, ut videtur, prior, omnium hominum omniumque aliarum artium et disciplinarum communis, alter natura posterior, proprius Philosophiae.*" Johannes de Raey to Wittich, August 1680, in *Cogitata de interpretatione*, Amsterdam: Apud Henricum Wetstenium, 1692, p. 661.

[39] "*Ac proinde intellectum et sermonem istum duplicem, quantum fieri potest, debere distingui, atque propter hanc potissimum distinctionem, Philosophiam ab omni alia arte et disciplina, censeri debere natura sua diversam esse.*" De Raey to Wittich, in *Cogitata*, p. 661.

believes.[40] But even if the first kind of language serves the demands of daily life, it is not for that reason a correct picture. As we will see in section 3.2 indeed, de Raey believes that an implicit reliance upon the pre-theoretical picture of the world that we develop on the basis of the impressions objects leave on our senses has served as a driving force behind some of the most problematic theories of matter to be found in scholastic metaphysics.

3.1. The Language of Daily Life

In his *Clavis philosophiae naturalis*, first published in 1654, de Raey argues that some aspects of the scholastic picture of reality and its vocabulary come naturally to us. We believe that magnets attract one another, that drugs cure our bodies, and that fire heats the substances around it. On the basis of this, we are naturally led to postulate powers and faculties in bodies that give rise to these activities:

> From our earliest years onwards, we have been assuming that, apart from the sensible qualities that affect our senses by themselves, we also cognize some *natural powers and faculties*, by means of which bodies either act upon themselves, or upon our bodies, or are capable of bringing about a change in some other body. Examples include gravity, levity, magnetic powers, and the faculties of many drugs, poisons, and other bodies. [41]

But from a philosophical point of view, de Raey claims, this picture of the world is incorrect. Bodies are inert, and do not initiate any motion. In fact, the only self-movers in the created universe are our souls, and to treat bodies

[40] On the letter to Wittich, see also Antonella Del Prete, "*Duplex Intellectus et Sermo Duplex*. Method and the Separation of Disciplines in Johannes de Raey," in Delpine Antoine-Mahut and Sophie Roux (eds.), *Physics and Metaphysics in Descartes and in his Reception*, London: Routledge, 2018, pp. 161–174, esp. pp. 162–163. Part of what de Raey is saying here, is that it would be a mistake to apply the language of philosophy to the domain of religious belief. The standards of clarity and rigour we apply in metaphysics must not be applied to religion. For discussion, see Theo Verbeek, "Les Cartésiens face à Spinoza: l'exemple de Johannes de Raey," in Paolo Cristofolini (ed.), *L'Hérésie Spinoziste. La discussion sur le* Tractatus Theologico-Politicus*, 1670-1677, et la réception immédiate du Spinozisme*, Amsterdam: APA-Holland University Press, 1991, pp. 77–88, and Andrea Strazzoni, "Johannes de Raey and the Cartesian Philosophy of Language," *Lias* 42 (2015), pp. 89–120, esp. pp. 93–97.

[41] "*Jam ab ineunte aetate assumptum a nobis fuit, praeter sensibiles qualitates sensus nostros per sese afficientes etiam naturales quasdam a nobis cognosci potentias facultatesque, quibus corpora vel in se ipsis operari, vel in corpore nostro, aut alio quovis mutationem quondam induci valeant; talesque sunt gravitas, levitas, vires magneticae et multae medicamentorum, venenorum aliorumque corporum facultates.*" Johannes de Raey, *Clavis philosophiae naturalis, seu introductio ad naturae contemplationem Aristotelico-Cartesiana*, Leiden: Elsevier, 1654, p. 12.

as substances endowed with powers or faculties to bring about certain effect is to project the nature of thinking substance onto extended substance.[42]

Yet the language of substances endowed with powers and faculties is not without use altogether. According to de Raey indeed, it has a clear enough meaning and function in daily life and practice:

> We believe that the signification of such nouns must not be removed from human speech, which we use in daily life, or in those disciplines which regard what is useful for that life, as if they were *insignificant words*, as Hobbes has it. Nor must they be substituted by others, as if instead of *faculty, power, action, life, soul*, we should rather speak of *motion, elementary matter, celestial globules*, *grooved corpuscles* and other new names, just because these ordinary words do not signify, or do not make known to listeners, that which a Philosopher would want them to signify or make known.[43]

For the daily business of medical practitioners, for instance, it is useful to be able to speak of the healing powers of certain substances, or the lack thereof. To be sure, references to healing powers could be rephrased away in a more scientifically apt way. But just as, in ordinary parlance, we gain nothing from substituting references to warmth by references to a certain kind of corpuscular motion, a medical practitioner who prescribes a certain drug would not become a better doctor if she refrained from saying that the drug has the power to alleviate pain and instead confined herself to claims about the micro-level corpuscular structure of the drug in question.[44] Indeed, de Raey goes so far as to say that if for scientific rigour's sake we decided to refrain from all references to powers and faculties, or warmth and coldness, there would be

> no speech among men, without which even those who claim that such words are insignificant, would not be able to speak, and would not speak. [45]

But if the language of powers and faculties may serve the needs of the medical practitioner in her daily business well enough, it must not be used when what we aim at is a scientific understanding of reality. In general, de

[42] See *Clavis*, pp. 68, 84 and 87.

[43] "*Putamusque horum nominum significationem, neque a humano sermone, quo vel in communi vita, vel in disciplinis utimur ad hujus vitae usum spectantibus removeri [ac si, ut loquitur Hobbesius,* voces insignificantes *sint] neque per substitutionem everti debere, ac si non amplius* facultas, vis, actio, Natura, vita, anima, *verum* motus, materia primi elementi, globuli coelestes, particulae striatae, *dicere, aliisve debeamus novis nominibus uti, propter hoc unum, quod usitata illa non significant, non faciant notum in extantibus, id quod in iis Philosophus desiderat.*" *Cogitata*, p. 213.

[44] See *Cogitata*, p. 210 on the substitution of '*calor*' for '*motus varius*.'

[45] "*Sine quo non consistit sermo inter homines, sine quo illi ipsi, qui contendunt has voces insignificantes esse, loqui non possunt, et non loquuntur.*" *Cogitata*, p. 213.

Raey believes that the notions we form prior to philosophical reflection are an important source of philosophical confusion. In particular, he holds that some of the most problematic scholastic accounts of the nature of matter are the result of an, arguably unintended, intrusion of our pre-theoretical picture of reality on the domain of metaphysics.

3.2. Prime Matter

First among the matter theories de Raey discusses is that of Thomas Aquinas.[46] According to Aquinas, prime matter is matter stripped of all form. Because it lacks a substantial form, it is not an element such as water or air, and because it lacks accidental forms, it is deprived of accidents such as colour or smell. Prime matter is bare matter, and as such it serves as the material ingredient out of which all concrete materials are made. Thus water is made out of prime matter that is determined to a certain nature by the form of water, and air is made out of prime matter that is determined to a certain nature by the form of air.

Because prime matter is indifferent to all forms and is just as receptive of the form of water as it is of any other form, it has what we may call a universal potency for form. But because it has no form in and of itself, this potency is in fact all that it has. More precisely, because it has no form in and of itself, prime matter in and of itself has no actuality. This follows, because a form according to Aquinas is a principle that causes something to be actually such or so, so that if matter lacks all form it cannot be actually anything. Thus Aquinas arrives at the conclusion that the material ingredient out of which all concrete bodies of the material world are made, is pure potency and nothing in actuality.

De Raey rejects this view as absurd. Matter construed as pure potency and as nothing in actuality cannot be the kind of stuff the concrete bodies we see around us are made of. But absurd as the Thomist account of prime matter may be, he also believes that its roots lie in the pre-theoretical way in which we have all grown up to understand the world:

> We can identify two causes of this error. The first consists in this prejudice, common to all men, according to which the subsistence and reality of bodies is deemed to depend on the forms and qualities that affect the senses to such a degree that, in a body stripped from these, nothing substantial remains.[47]

[46] For Aquinas on prime matter, see for instance section 2 of *De principiis naturae*, in Thomas Aquinas, *Opera Omnia*, editio Leonina, Rome: Commissio Leonina, 1882–, vol. 43, pp. 40–41.

[47] "*Duaeque notari possunt hujus erroris causae: prima consistit in communi illo omnium hominum praejudicio, quo corporum subsistentiam ac realitatem sic pendere putant a formis atque in sensum occurentibus qualitatibus, ut iis sepositis nihil substantiale sit reliquum.*" *Clavis*, p. 55.

One of the main prejudices we develop as children, according to de Raey, is the prejudice that the senses are reliable guides to the way the world is. One product of this prejudice is a picture of the world that is drawn by the senses to such an extent that many of us find it hard to see how something that cannot be sensed can be actual at all:[48]

> That which is stripped of all sensible form and quality, and that also, which we conceive to be in bodies apart from those, we easily take to be nothing at all.[49]

The second cause that underlies the Thomist conception of matter, we are told in the *Clavis*, is the idea that, in order to be able to account for the miracle of the Eucharist, the quantity of a body must be treated as an accidental form that could be separated from it.[50] According to de Raey, this conception of quantity as an accidental form implies that, in and of itself, matter is not extended. But unextended matter, according to de Raey, is nothing at all. The moment we allow that quantity is an accidental form of matter, then, we commit ourselves to the Thomist view according to which, in and of itself, matter is nothing.

Neither of these diagnoses seems particularly successful. The first ascribes to Aquinas an inference from

prime matter has no sensible form

to

prime matter is nothing in actuality.

This inference is obviously problematic. In reality, however, Aquinas was making an inference from

prime matter has no form at all

to the non-actuality of prime matter. To be sure, much more could be said about the inference here and the premises on which it relies. But it seems fair to grant that the step from 'prime matter has no form at all' to the non-actuality of matter is relevantly smaller than the leap de Raey has Aquinas make.

As for the second diagnosis, it seems that the inference from 'matter in and of itself is unextended' to 'matter in and of itself is nothing' relies on

[48] See *Clavis*, p. 52 for de Raey on the *primae aetatis praejudicia*. The reference to the prejudices of childhood is of course reminiscent of Descartes, who in the *Principles* had identified the juvenile prejudice that the senses are a reliable guide to reality as one of the main sources of error in philosophy. See *Oeuvres*, vol. 8A, p. 35 / *Philosophical Writings,* vol. 1, pp. 218–219.

[49] "*Id, quod omni sensibili forma, et qualitate revera destitutum adeoque a sensibilibus corporibus separatum est; ita id etiam, quod sine iis praecise consideratur in corporibus, facile pro nihilo habemus.*" *Clavis*, p. 52.

[50] See *Clavis*, p. 55.

the implicit premise that quantity is essential to matter, so that unquantified matter could not exist. But although it is clear why a Cartesian such as de Raey would accept this implicit premise, it is not at all clear that his scholastic interlocutors would. On the contrary, it seems that no one who thinks that quantity is an accidental form of matter should accept his inference here.

But in spite of these flaws, his analysis of the Thomist conception of matter does illustrate how, according to de Raey, the picture of reality that we have developed from our earliest days onwards, when taken to the extreme, provides the breeding ground for what he takes to be absurd positions in philosophy, such as that prime matter is pure potency, and that the material ingredient of natural materials, accordingly, is nothing in actuality. How should we avoid this kind of view?

4. *Matter as Extension*

The first alternative to Aquinas's theory of matter de Raey discusses is that of Avicenna. According to Avicenna, prime matter is eternally endowed with a generic form of corporeity. It is contracted to the nature of some determinate kind of matter by a subsequent, specific, form:

> A body, insofar as it is a body, has a principle that is its matter and a principle that is its form – whether it be the corporeal form common to all bodies, or some specific form from the range of bodily forms.[51]

Although the exact role and function of this corporeal form continues to be the matter of debate among scholars, Avicenna seems to want to say that the corporeal form of matter is what gives it extension in three dimensions.[52] Thus even though prime matter is deprived of all specific forms (it lacks the nature of any concrete material), it does not follow that it has no actuality at all, as it did in Aquinas. After all, even though it may not have the form of water, air, or any other material, it still has actual extension in three dimensions in virtue of its corporeal form.

But in spite of the differences between Aquinas and Avicenna on prime matter, de Raey believes that the two are in agreement on a fundamental point. They agree that, unless something is superadded to matter, matter will lack all actuality. To be sure, they disagree over the nature of that something.

[51] "*Ponat ergo naturalis quod corpus, ex eo quod est corpus, habet principium quod est hyle et principium quod est forma, sive forma corporalis communis, sive forma specialis aliqua ex formis corporum.*" Avicenna, *Liber primus naturalium. Tractatus primus de causis et principiis naturalium*, ed. Simone Van Riet, Louvain: Peeters / Leiden: Brill, 1992, tr. 1, cap. 2 (p. 21).

[52] For discussion, see Andreas Lammers, *The Elements of Avicenna's Physics. Greek Sources and Arabic Innovations*, Berlin: De Gruyter, 2018, pp. 122–132.

According to Aquinas, a specific form that contracts matter to a material such as water will give it the actuality it lacks. According to Avicenna, a generic form of corporeity will do the job. But in the final analysis, the two agree that, in and of itself, matter lacks all actuality. And this, as we have already seen, is a view that de Raey believes can be traced back to a juvenile picture of reality painted by the senses.[53]

The scholastic philosopher to have come closest to the truth about the nature of matter, according to de Raey, was Jacob Zabarella.[54] In particular, de Raey applauded Zabarella for his rejection of Avicenna's form of corporeity. According to Zabarella, such a form would turn all other forms into accidental forms:

> If there were such an eternal form in prime matter, it would follow that no other forms except accidental ones could be received in it. For by means of such an eternal form, matter would become a being in actuality. But no substantial form can be added to a being in actuality, but only accidental ones.[55]

If a portion of matter is endowed with the form of corporeity, it will thereby become an actual body. Now according to Zabarella, all forms that are added to a body that is already actual are accidental forms. Hence if the form of gold is added to our actual body to make it gold rather than any other kind of matter, this form will count as an accidental form as well. But this seems wrong. The form of gold defines our body as the kind of body it is. Hence it must count as a substantial form, not as a mere accident.

A further problem with Avicenna's form of corporeity, according to Zabarella, is that it failed to do precisely what forms were most needed for: to discriminate between various species of material substances:

> Every form that cannot perform the office of a form, is posted in vain. But this form of corporeity is such. For the office of a form is to restrict matter, and to confine it to a certain nature. This form, however, brings about no such confinement of matter.[56]

[53] On Avicenna and 'the vulgar' on matter, see *Clavis*, p. 56.

[54] On Zabarella on matter, see Pasnau, *Metaphysical Themes*, pp. 66–71.

[55] "*Si in prima materia esset haec forma aeterna, sequeretur, non posse ullam aliam formam in materia recipi, sed sola accidentia, quoniam per hanc formam materia fieret ens actu; enti autem actu non potest aliqua substantialis forma advenire, sed accidentalia tantummodo.*" Jacob Zabarella, *De rebus naturalibus libri XXX*, Frankfurt: Sumptibus Haeredum Lazari Zetzneri, 1617, p. 208.

[56] "*Vana est omnis forma, quae formae officio non fungatur, hujusmodi autem est haec forma corporis; etenim quum formae officium sit materiam restringere, et coarctare, haec tamen nullam materiae coarctationem facit.*" *De rebus naturalibus*, p. 210.

Zabarella's own view is that matter is by its own nature extended in three dimensions. As de Raey summarizes:

> Matter, in virtue of its own nature, is a substantial body subject to three dimensions.[57]

Instead of saying that matter is extended by nature, Zabarella also says that it has quantity by nature. But even though matter has quantity by nature, Zabarella does not believe that quantity is what constitutes the essence of matter. Rather, it is a feature that follows from the essence of matter, just as the capacity for laughter is a feature that does not constitute but follow from the essence of man:

> Quantity follows from the nature of matter, in the way in which the capacity for laughter follows from the nature of man. And quantity is not separable from matter, but it does not constitute the essence of matter either.[58]

But de Raey does not see the need for this distinction between the essence of matter and its extension. If Zabarella believes that quantity flows from the essence of matter, de Raey believes, he could just as well have said that quantity is what constitutes the essence of matter.[59] Indeed this is the view of matter that de Raey, following Descartes, adopts himself.

At this point, we can also see how their different methodological points of departure cause authors like Digby and de Raey to move in opposite directions. According to Digby, philosophy must remain grounded in the pre-theoretical picture of reality common to all men. Since this picture tells us that bodies have a variable extension, we must allow for some kind of distinction between material substance and extension in philosophy. But where Digby aims to build upon our pre-theoretical picture of reality, de Raey aims to replace it. In the ontology of matter, this requires moving from Aquinas and Avicenna to Zabarella and eventually Descartes, and towards the denial of any distinction between material substance and extension. To be sure, de Raey knew just as well as Digby that we often speak as if one and the same portion of matter becomes larger or smaller. But where Digby takes this fact about ordinary parlance to put restraints on what can count as a viable theory of matter, de Raey would take it to illustrate once more how ways of speaking

[57] "*Secundam propriam suam natura est corpus substantiale simplex trinae dimensioni subjectum.*" *Clavis*, 56.

[58] "*Essentiam enim materiae insequitur quantitas, sicut essentiam hominis risibilitas, quare nec separabilis est a materia, nec materia est essentialiter quantitas.*" *De rebus naturalibus*, p. 204.

[59] *Clavis*, p. 57.

that serve the purposes of everyday practice well enough will put us on the wrong track in philosophy.

5. *Conclusion*

How should philosophy relate to our pre-theoretical picture of reality? In this paper, we have encountered two philosophers who were willing to appeal to the former to lay out and develop their views on the nature of material reality. But in spite of their agreement on this general point of methodology, Digby and Soto disagreed on what it is that the common way of conceiving of the world tells us. According to Soto, it tells us that an apple and its colour differ as two *res*. According to Digby, it tells us that the colour of the apple is nothing over and above the apple itself insofar as it acts upon the human senses in a certain way. To be sure, there is much more that could be said about the claims and arguments Soto and Digby adduce in favour of their views. But if we zoom out a bit from the details of their accounts, we see that they serve as a reminder that it is not at all clear that our pre-theoretical picture of the world carries any clear messages about the nature of material substance or the ontological status of accidents. This brings us to the case of de Raey, who believed that the picture of reality we naturally develop from our early days onwards and the language associated with it have their use in domains such as law and medical practice, but are a poor guide to the ontology of the material world. If we are to arrive at a proper understanding of the nature of matter, for instance, we must move beyond the picture of reality we naturally develop from our early days onwards in order to come to see that the nature of matter consists in bare extension.

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